

ADVANCED

Stage 6

Functions (Adv), F2 Graphing (Adv)

Transformations (Y12)

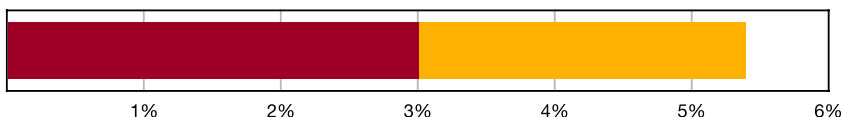
Non-Calculus Graphing (Y12)

Teacher: Kristy Chen

Exam Equivalent Time: 180 minutes (based on allocation of 1.5 minutes per mark)



F2 Graphing



*SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.

HISTORICAL CONTRIBUTION

- F2 Graphing Techniques* has contributed an average of 5.4% per Advanced exam since the new syllabus was introduced in 2020.
- We have split the topic into 2 categories for analysis purposes: *1-Transformations (3.0%)* and *2-Non-Calculus Graphing (2.4%)*.
- This analysis looks at *Transformations*.

HSC ANALYSIS - What to expect and common pitfalls

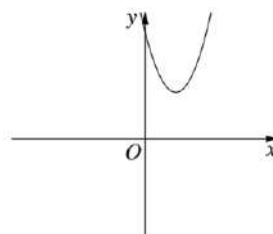
- Transformations (3.0%)* represents new syllabus content that explicitly looks at translations and dilations of several function types.
- Absolute value* graph transformations were last examined in *2023 Adv 27* with a 5-mark allocation. A "must revise" question where part (b) produced a 23% mean mark.
- Questions typically involve combinations of translations, dilations and/or reflections. Mean marks have hovered around 50% and revision attention is required (see *2022 Adv 19* and *2021 Adv 21*).
- The NESA sample HSC exam has informed *F2 EQ-Bank* questions which are important revision.
- We note that Trig transformations, which we regard as an extremely important transformation sub-topic, are covered separately under *T3 Trig Graphs*.

Questions

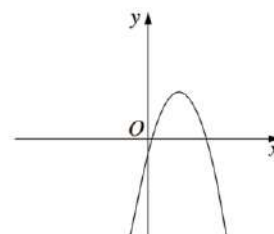
1. Functions, 2ADV F2 2016 HSC 3 MC

Which diagram best shows the graph of the parabola $y = 3 - (x - 2)^2$?

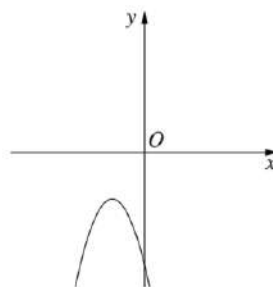
(A)



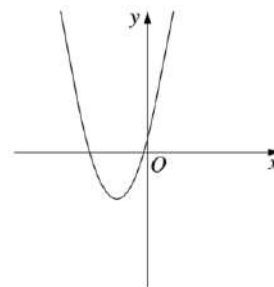
(B)



(C)



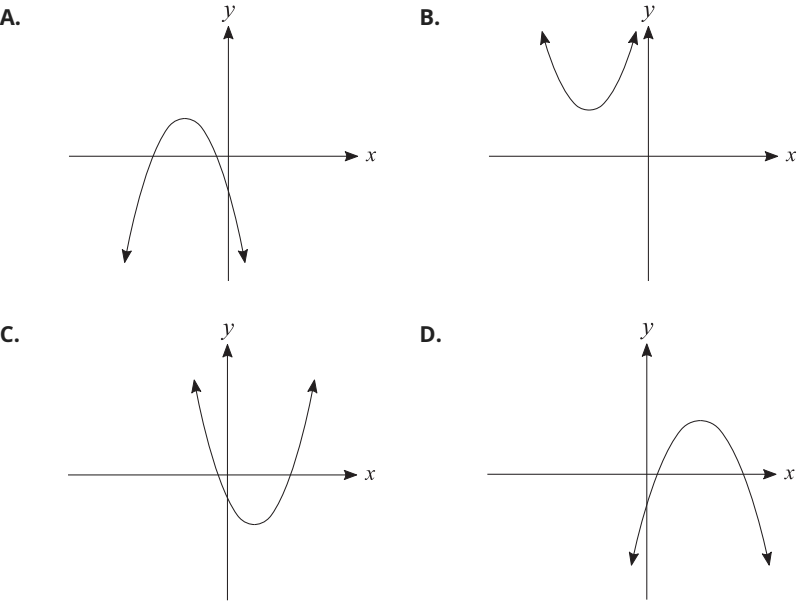
(D)



2. Functions, 2ADV F2 EQ-Bank 2 MC

Which diagram best shows the graph

$y = 1 - 2(x + 1)^2$



3. Functions, 2ADV F2 2020 HSC 2 MC

The function $f(x) = x^3$ is transformed to $g(x) = (x - 2)^3 + 5$ by a horizontal translation of 2 units followed by a vertical translation of 5 units.

Which row of the table shows the directions of the translations?

	Horizontal translation of 2 units	Vertical translation of 5 units
A.	Left	Up
B.	Right	Up
C.	Left	Down
D.	Right	Down

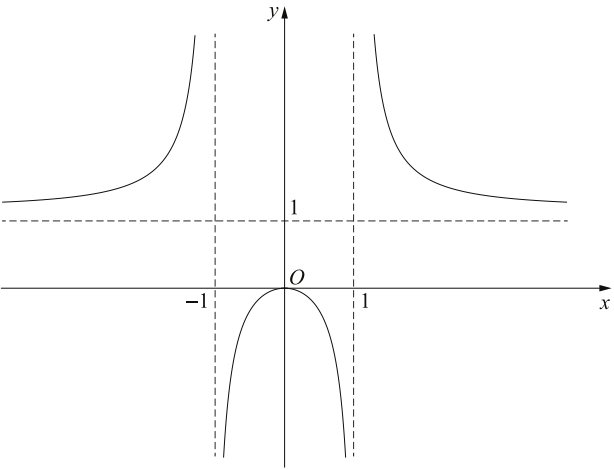
4. Functions, 2ADV F2 SM-Bank 9 MC

The graph of the function $f(x) = \frac{3x + 2}{5 - x}$, has asymptotes at

- A. $x = -5, y = \frac{3}{2}$
- B. $x = \frac{2}{3}, y = -3$
- C. $x = 5, y = 3$
- D. $x = 5, y = -3$

5. Functions, 2ADV' F2 2019 HSC 4 MC

The diagram shows the graph of $y = f(x)$.



Which equation best describes the graph?

- A. $y = \frac{x}{x^2 - 1}$
- B. $y = \frac{x^2}{x^2 - 1}$
- C. $y = \frac{x}{1 - x^2}$
- D. $y = \frac{x^2}{1 - x^2}$

6. Functions, 2ADV F2 SM-Bank 5 MC

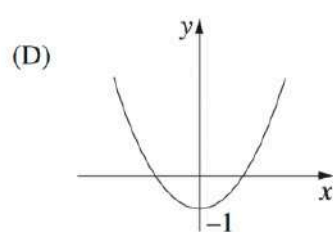
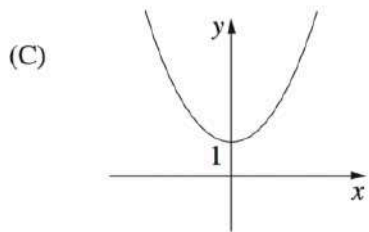
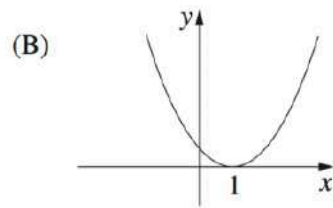
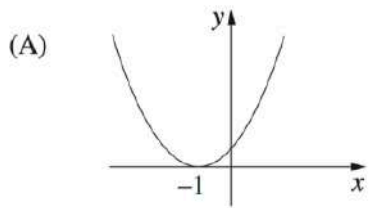
The point $P(4, -3)$ lies on the graph of a function $f(x)$. The graph of $f(x)$ is translated four units vertically up and then reflected in the y -axis.

The coordinates of the final image of P are

- A. $(-4, 1)$
- B. $(-4, 3)$
- C. $(0, -3)$
- D. $(4, -6)$

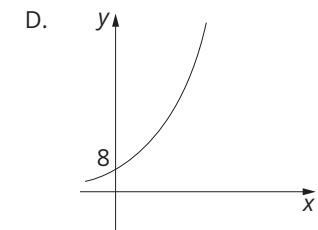
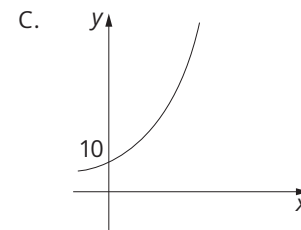
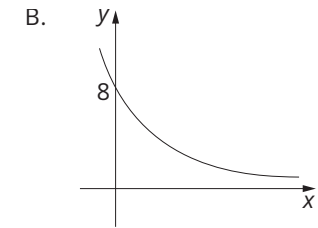
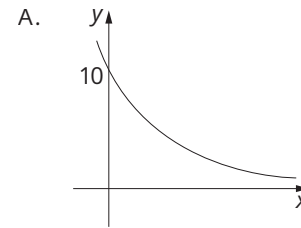
7. Functions, 2ADV F2 2014 HSC 2 MC

Which graph best represents $y = (x - 1)^2$?



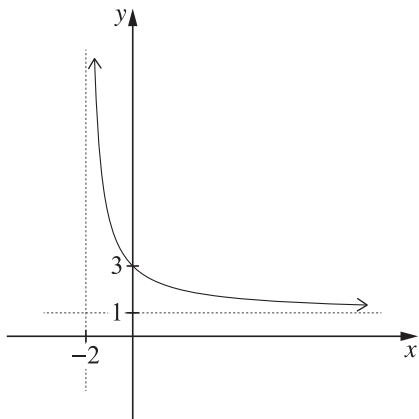
8. Functions, 2ADV F2 2021 HSC 5 MC

Which of the following best represents the graph of $y = 10(0.8)^x$?



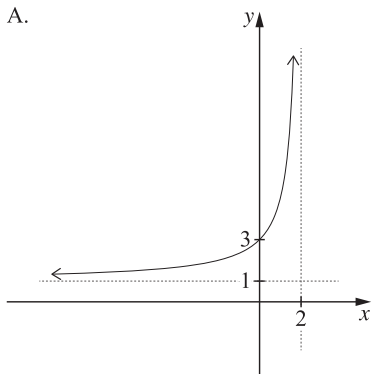
9. Functions, 2ADV F2 2025 HSC 6 MC

The graph of $y = f(x)$ is shown.

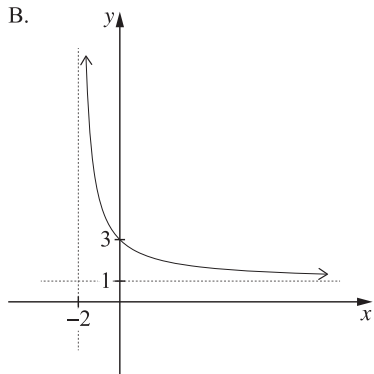


Which of the following is the graph of $y = -f(-x)$?

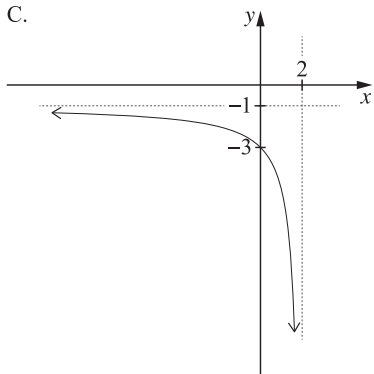
A.



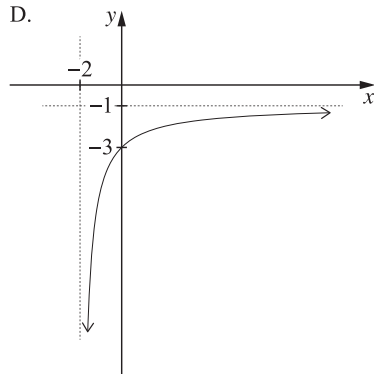
B.



C.



D.



10. Functions, 2ADV F2 EQ-Bank 3 MC

The graph $y = \frac{2}{x-2}$ undergoes the following transformations:

- translated 3 units to the left
- dilated vertically by a factor of 2

Determine which of the following is the new function.

A. $2y = \frac{2}{x+1}$

B. $\frac{y}{2} = \frac{2}{x-5}$

C. $2y = \frac{2}{x-5}$

D. $\frac{y}{2} = \frac{2}{x+1}$

11. Functions, 2ADV F2 SM-Bank 6 MC

The graph of a function $f(x)$ is obtained from the graph of the function $g(x) = \sqrt{2x-5}$ by a reflection in the x -axis followed by a dilation from the y -axis by a factor of $\frac{1}{2}$.

Which one of the following is the function $f(x)$?

A. $f(x) = \sqrt{5-4x}$

B. $f(x) = -\sqrt{x-5}$

C. $f(x) = \sqrt{x+5}$

D. $f(x) = -\sqrt{4x-5}$

12. Functions, 2ADV F2 SM-Bank 4 MC

The graph of the function $f(x) = 3x^{\frac{5}{2}}$ is reflected in the x -axis and then translated 3 units to the right and 4 units down.

The equation of the new graph is

A. $y = 3(x-3)^{\frac{5}{2}} + 4$

B. $y = -3(x-3)^{\frac{5}{2}} - 4$

C. $y = -3(x+3)^{\frac{5}{2}} - 1$

D. $y = -3(x-4)^{\frac{5}{2}} + 3$

13. Functions, 2ADV F2 SM-Bank 10 MC

The graph of the function $f(x) = \frac{x-3}{2-x}$ has asymptotes

- A. $x = 3,$ $y = 2$
- B. $x = -2,$ $y = 1$
- C. $x = 2,$ $y = -1$
- D. $x = 2,$ $y = 1$

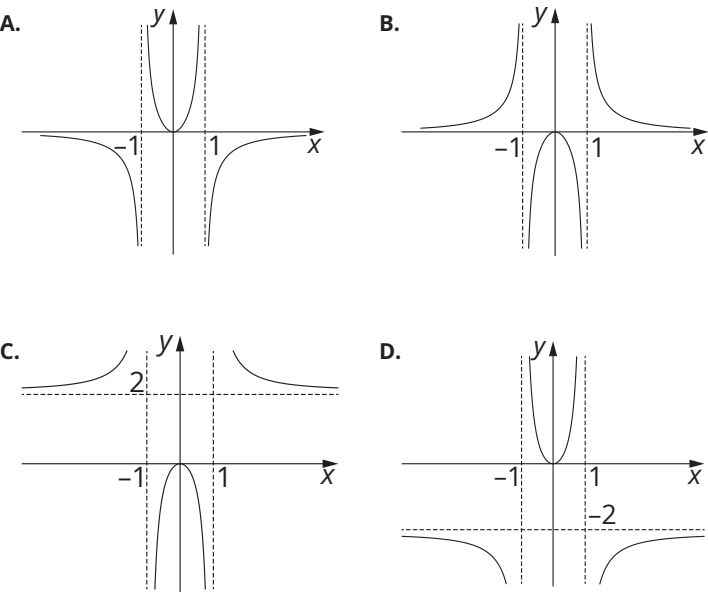
14. Functions, 2ADV' F2 2015 HSC 5 MC

What are the asymptotes of $y = \frac{3x}{(x+1)(x+2)}$

- A. $y = 0,$ $x = -1,$ $x = -2$
- B. $y = 0,$ $x = 1,$ $x = 2$
- C. $y = 3,$ $x = -1,$ $x = -2$
- D. $y = 3,$ $x = 1,$ $x = 2$

15. Functions, 2ADV' F2 2017 HSC 5 MC

Which graph best represents the function $y = \frac{2x^2}{1-x^2}$?



16. Functions, 2ADV F2 2024 HSC 4 MC

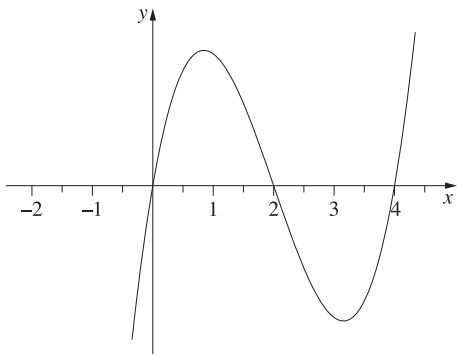
The parabola $y = (x-3)^2 - 2$ is reflected about the y -axis. This is then reflected about the x -axis.

What is the equation of the resulting parabola?

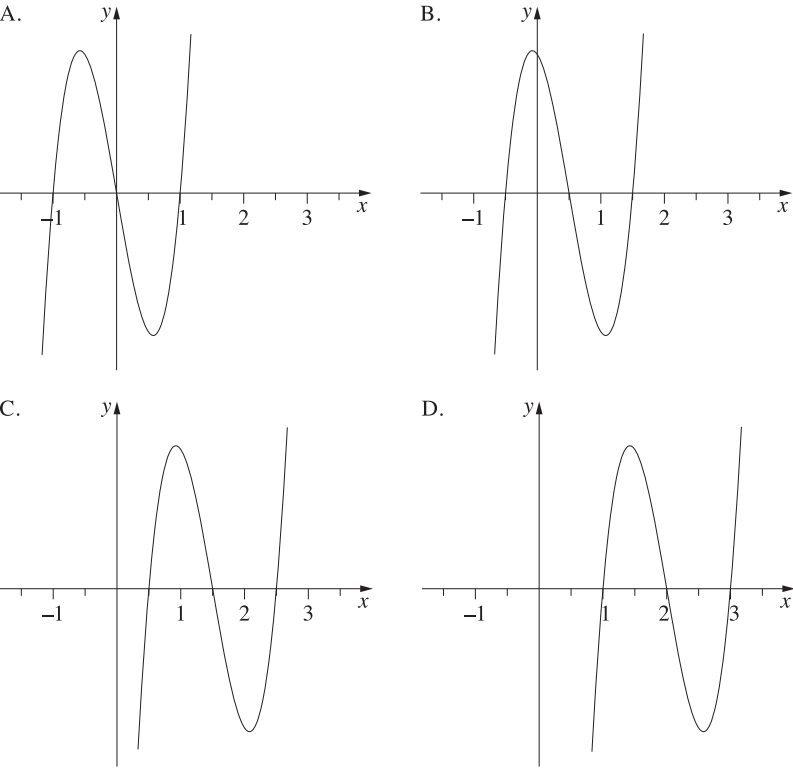
- A. $y = (x+3)^2 + 2$
- B. $y = (x-3)^2 + 2$
- C. $y = -(x+3)^2 + 2$
- D. $y = -(x-3)^2 + 2$

17. Functions, 2ADV F2 2024 HSC 7 MC

The diagram shows the graph $y = f(x)$.

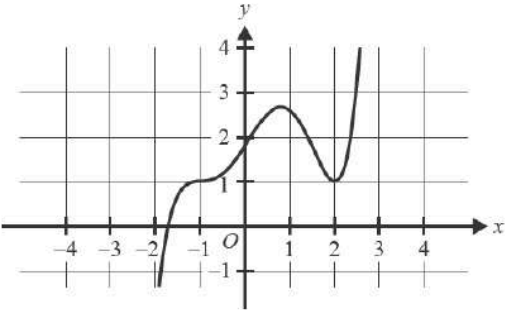


Which of the following best represents the graph $y = f(2x - 1)$?

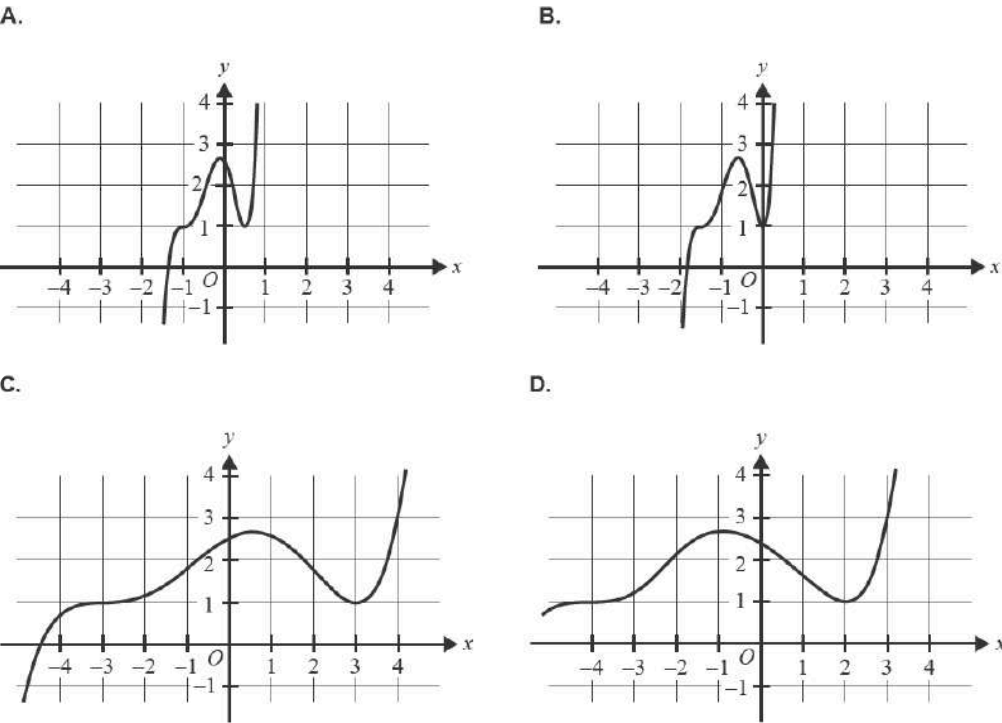


18. Functions, 2ADV F2 2024 MET2 12 MC

The graph of $y = f(x)$ is shown below.



Which of the following options best represents the graph of $y = f(2x + 1)$?



19. Functions, 2ADV F2 2022 SPEC2 3 MC

The graph of $y = \frac{x^2 + 2x + c}{x^2 - 4}$, where $c \in \mathbb{R}$, will always have

- A. two vertical asymptotes and one horizontal asymptote.
- B. a vertical asymptote with equation $x = -2$ and one horizontal asymptote with equation $y = 1$.
- C. one horizontal asymptote with equation $y = 1$ and only one vertical asymptote with equation $x = 2$.
- D. a horizontal asymptote with equation $y = 1$ and at least one vertical asymptote.

20. Functions, 2ADV F2 SM-Bank 7 MC

The point $A(3, 2)$ lies on the graph of the function $f(x)$. A transformation maps the graph of $f(x)$ to the graph of $g(x)$,

where $g(x) = \frac{1}{2}f(x - 1)$. The same transformation maps the point A to the point P .

The coordinates of the point P are

- A. (2, 1)
- B. (2, 4)
- C. (4, 1)
- D. (4, 2)

21. Functions, 2ADV F2 SM-Bank 8 MC

The transformation that maps the graph of $y = \sqrt{8x^3 + 1}$ onto the graph of $y = \sqrt{x^3 + 1}$ is a

- A. dilation by a factor of 2 from the y -axis.
- B. dilation by a factor of 2 from the x -axis.
- C. dilation by a factor of $\frac{1}{2}$ from the x -axis.
- D. dilation by a factor of $\frac{1}{2}$ from the y -axis.

22. Functions, 2ADV F2 2006 HSC 1c

Sketch the graph of $y = |x + 4|$. (2 marks)

23. Functions, 2ADV F2 SM-Bank 1

- i. Draw the graph $y = \ln x$. (1 mark)
- ii. Explain how the above graph can be transformed to produce the graph

$$y = 3\ln(x + 2)$$

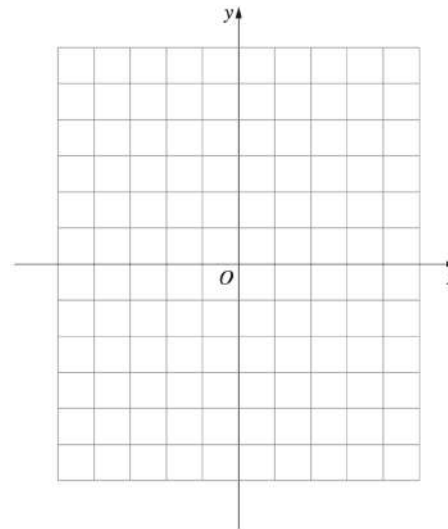
and sketch the graph, clearly identifying all intercepts. (3 marks)

24. Functions, 2ADV F2 SM-Bank 35

- i. Sketch the function $y = f(x)$ where $f(x) = (x - 1)^3$ on a number plane, labelling all intercepts. (1 mark)
- ii. On the same graph, sketch $y = -f(-x)$. Label all intercepts. (2 marks)

25. Functions, 2ADV F2 2023 HSC 19

- a. Sketch the graphs of the functions $f(x) = x - 1$ and $g(x) = (1 - x)(3 + x)$ showing the x -intercepts. (2 marks)



- b. Hence, or otherwise, solve the inequality $x - 1 < (1 - x)(3 + x)$. (2 marks)

26. Functions, 2ADV F2 EQ-Bank 16

$y = -\frac{(x + 2)^4}{3}$ has been produced by three successive transformations: a translation, a dilation and then a reflection.

- i. Describe each transformation and state the equation of the graph after each transformation. (2 marks)
- ii. Sketch the graph. (1 mark)

27. Functions, 2ADV F2 EQ-Bank 9

Consider the function $f(x) = \frac{1}{4x - 1}$.

- i. Find the domain of $f(x)$. (1 mark)
- ii. Sketch $f(x)$, showing all asymptotes and intercepts? (2 marks)

28. Functions, 2ADV' F2 2012 HSC 13b

- i. Find the horizontal asymptote of the graph $y = \frac{2x^2}{x^2 + 9}$. (1 mark)
- ii. Without the use of calculus, sketch the graph $y = \frac{2x^2}{x^2 + 9}$, showing the asymptote found in part (i). (2 marks)

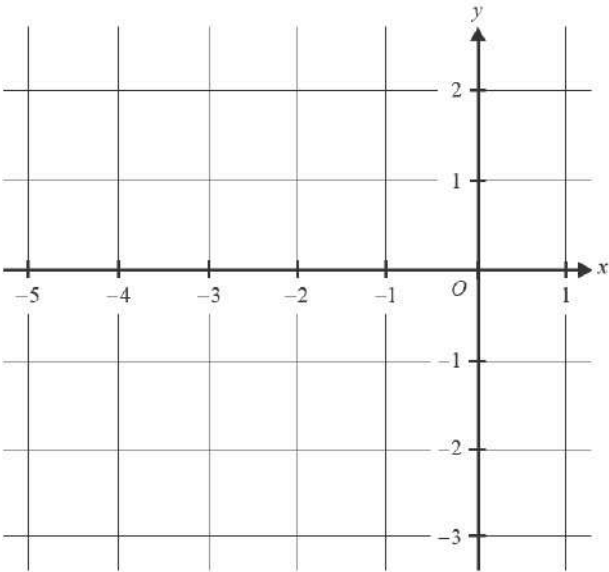
29. Functions, 2ADV F2 2021 HSC 19

Without using calculus, sketch the graph of $y = 2 + \frac{1}{x + 4}$, showing the asymptotes and the x and y intercepts. (3 marks)

30. Functions, 2ADV F2 2024 MET1 3*

Let $f(x) = \frac{1}{(x + 3)^2} - 2$.

On the axes below, sketch the graph of $y = f(x)$, labelling all asymptotes with their equations and axis intercepts with their coordinates. (4 marks)



31. Functions, 2ADV F2 2025 HSC 30

The parabola with equation $y = (x - 1)(x - 5)$ is translated both horizontally to the right and vertically up by k units, where k is positive.

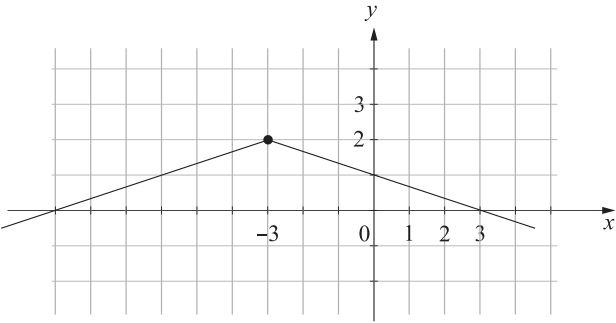
The translated parabola passes through the point $(6, 11)$.

Find the value of k . (3 marks)

32. Functions, 2ADV F2 EQ-Bank 1

The function $f(x) = |x|$ is transformed and the equation of the new function is $y = kf(x + b) + c$.

The graph of the new function is shown below.



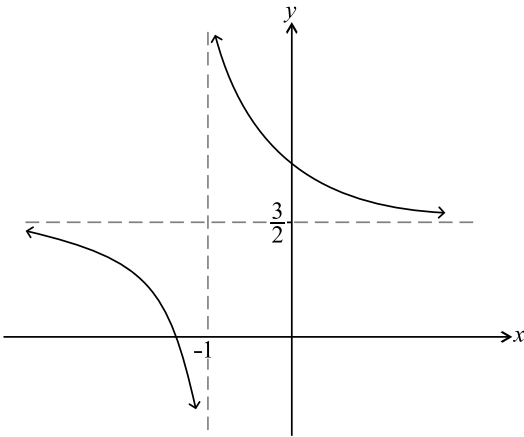
What are the values of k , b and c . (2 marks)

33. Functions, 2ADV F2 EQ-Bank 14

List a set of transformations that, when applied in order, would transform $y = x^2$ to the graph with equation $y = 1 - 6x - x^2$. (3 marks)

34. Functions, 2ADV F2 EQ-Bank 2

A graph of the hyperbola $y = \frac{1}{x + p} + q$ is shown, where p and q are constants.



Find the values of p and q and hence the graphs x -axis intercept. (2 marks)

35. Functions, 2ADV F2 SM-Bank 16

Let $f(x) = x^2 - 4$

Let the graph of $g(x)$ be a transformation of the graph of $f(x)$ where the transformations have been applied in the following order:

- dilation by a factor of $\frac{1}{2}$ from the vertical axis (parallel to the horizontal axis)
- translation by two units to the right (in the direction of the positive horizontal axis)

Find $g(x)$ and the coordinates of the horizontal axis intercepts of the graph of $g(x)$. (3 marks)

36. Functions, 2ADV F2 SM-Bank 13

i. Show that the function $y = \frac{1 - e^x}{1 + e^x}$ is an odd function? (1 mark)

ii. Sketch $y = \frac{1 - e^x}{1 + e^x}$, labelling all intercepts and asymptotes. (2 marks)

37. Functions, 2ADV F2 SM-Bank 2

Sketch the graph $y = \log_2(x - 3)$.

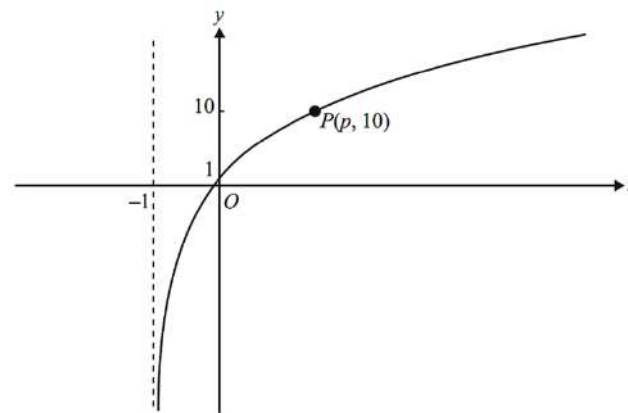
Show all asymptotes and state its domain and range. (3 marks)

38. Functions, 2ADV F2 SM-Bank 3

The diagram below shows part of the graph of the function with rule

$$f(x) = k \log_e(x + a) + c, \text{ where } k, a \text{ and } c \text{ are real constants.}$$

- The graph has a vertical asymptote with equation $x = -1$.
- The graph has a y -axis intercept at 1.
- The point P on the graph has coordinates $(p, 10)$, where p is another real constant.



i. State the value of a . (1 mark)

ii. Find the value of c . (1 mark)

iii. Show that $k = \frac{9}{\log_e(p + 1)}$. (2 marks)

39. Functions, 2ADV F2 SM-Bank 10

Consider the function $f(x) = \frac{x}{4 - x^2}$.

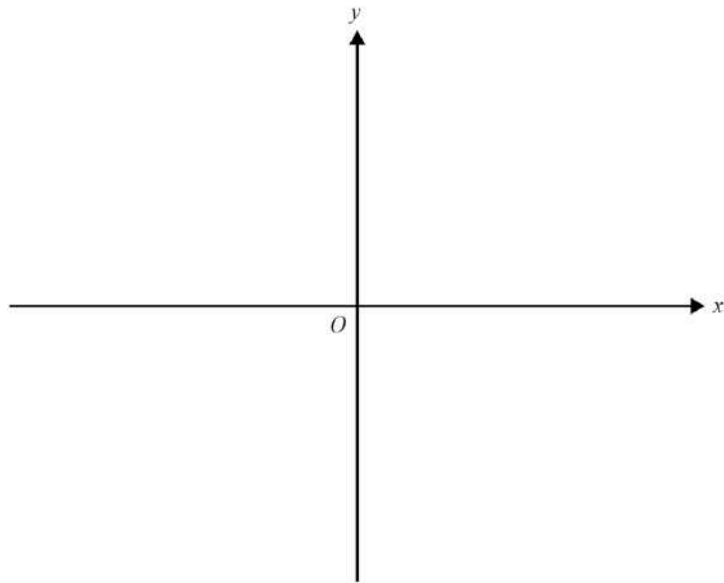
i. Identify the domain of $f(x)$. (1 mark)

ii. Sketch the graph $y = f(x)$, showing all intercepts and asymptotes. (3 marks)

40. Functions, 2ADV F2 EQ-Bank 11

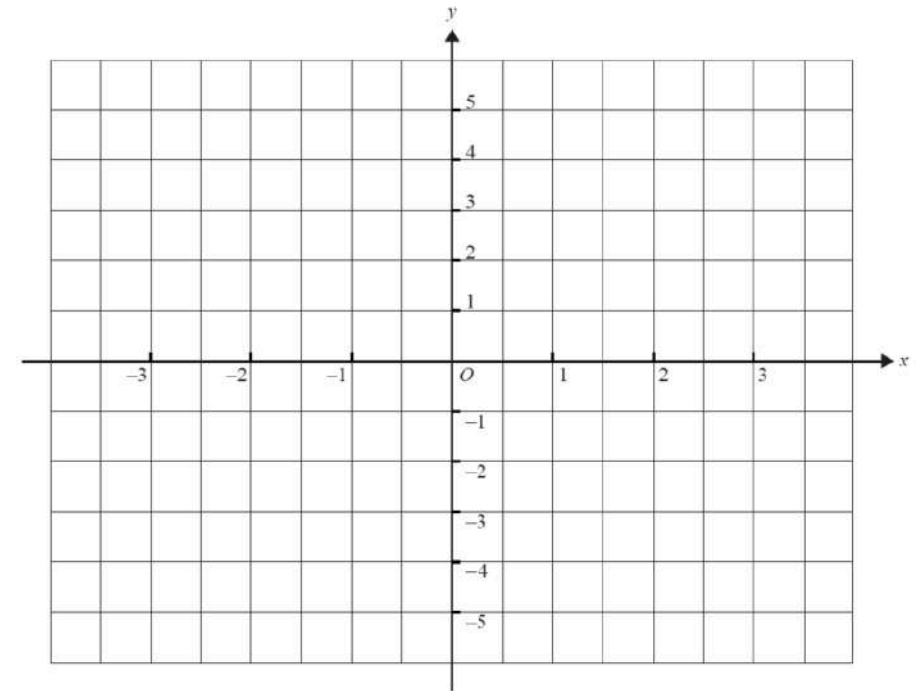
On the axes below, sketch the graph of $f(x) = \frac{2x-2}{x+1}$.

Label all axis intercepts. Label each asymptote with its equation. **(4 marks)**



41. Functions, 2ADV F2 SM-Bank 12

Sketch the graph of $f(x) = \frac{2x+1}{x-1}$. Label the axis intercepts with their coordinates and label any asymptotes with the appropriate equation. **(4 marks)**



42. Functions, 2ADV' F2 2007 HSC 3b

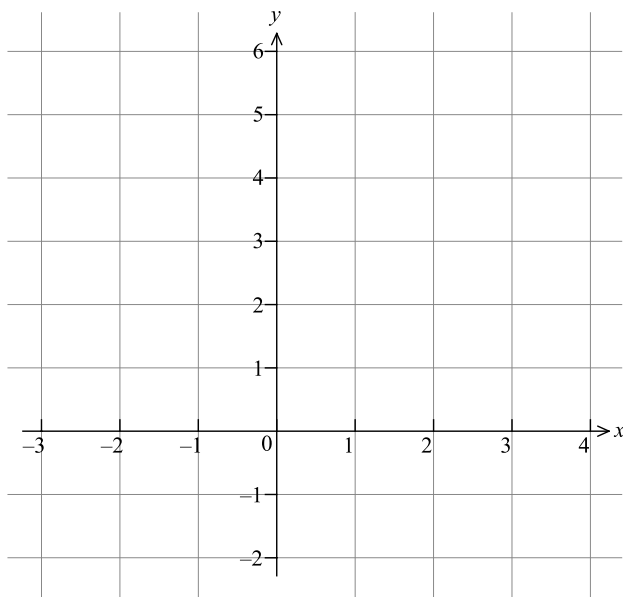
i. Find the vertical and horizontal asymptotes of the hyperbola

$y = \frac{x-2}{x-4}$ and hence sketch the graph of $y = \frac{x-2}{x-4}$. **(3 marks)**

ii. Hence, or otherwise, find the values of x for which $\frac{x-2}{x-4} \leq 3$. **(2 marks)**

43. Functions, 2ADV F2 2023 MET1 3

- a. Sketch the graph of $f(x) = 2 - \frac{3}{x-1}$ on the axes below, labelling all asymptotes with their equation and axial intercepts with their coordinates. (3 marks)



- b. Find the values of x for which $f(x) \leq 1$. (1 mark)

44. Functions, 2ADV F2 EQ-Bank 13

The curve $y = kx^2 + c$ is subject to the following transformations

- Translated 2 units in the positive x -direction
- Dilated in the positive y -direction by a factor of 4
- Reflected in the y -axis

The final equation of the curve is $y = 8x^2 + 32x - 8$.

- Find the equation of the graph after the dilation. (1 mark)
- Find the values of k and c . (2 marks)

45. Functions, 2ADV F2 EQ-Bank 2

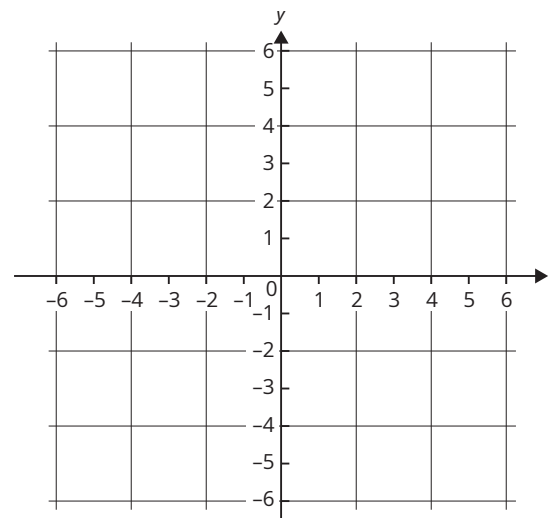
The curve $f(x) = x^2$ is transformed to $g(x) = 3f[2(x+2)]$

- Write the equation of $g(x)$. (1 mark)
- $P(-3, 9)$ lies on $f(x) = x^2$

Determine the corresponding co-ordinates of P on the curve $g(x)$. (2 marks)

46. Functions, 2ADV F2 SM-Bank 20

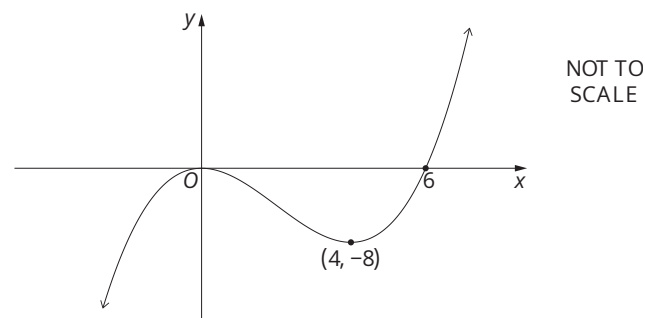
- a. Sketch the graph of $y = 1 - \frac{2}{x-2}$ on the axes below. Label asymptotes with their equations and axis intercepts with their coordinates. (3 marks)



- b. Find the values of x for which $1 - \frac{2}{x-2} \geq 3$. (1 mark)

47. Functions, 2ADV F2 2021 HSC 21

Consider the graph of $y = f(x)$ as shown.



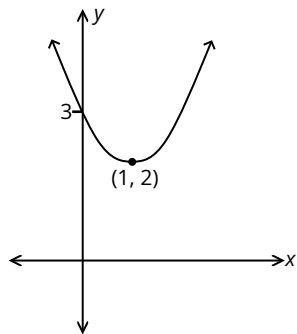
Sketch the graph of $y = 4f(2x)$ showing the x -intercepts and the coordinates of the turning points. (2 marks)

48. Functions, 2ADV F2 SM-Bank 2

$f(x) = (x + 2)^2$ is transformed and the equation of the new function is in the form

$$y = kf(x + a) + c, \text{ where } k, a \text{ and } c \text{ are constants.}$$

The graph of the transformed function is shown below.



Determine the values of k , a and c . (3 marks)

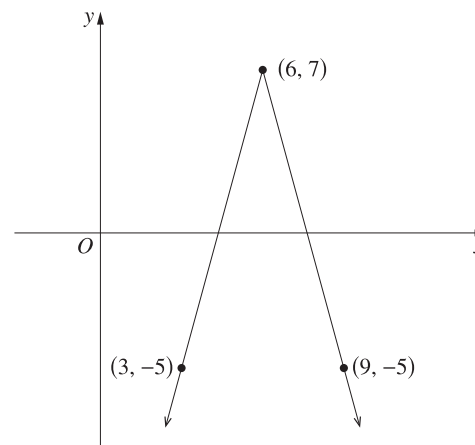
49. Functions, 2ADV F2 2022 HSC 19

The graph of the function $f(x) = x^2$ is translated m units to the right, dilated vertically by a scale factor of k and then translated 5 units down. The equation of the transformed function is $g(x) = 3x^2 - 12x + 7$.

Find the values of m and k . (3 marks)

50. Functions, 2ADV F2 2023 HSC 27

The graph of $y = f(x)$, where $f(x) = a|x - b| + c$, passes through the points $(3, -5)$, $(6, 7)$ and $(9, -5)$ as shown in the diagram.



a. Find the values of a , b and c . (3 marks)

b. The line $y = mx$ cuts the graph of $y = f(x)$ in two distinct places.

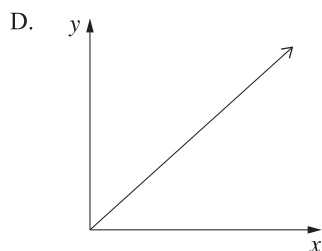
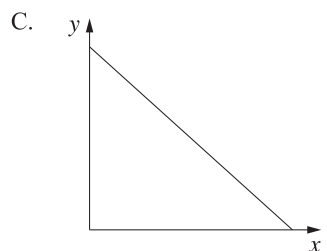
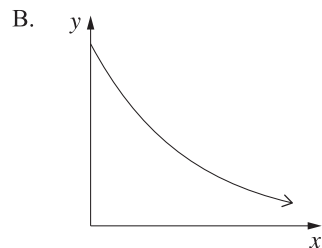
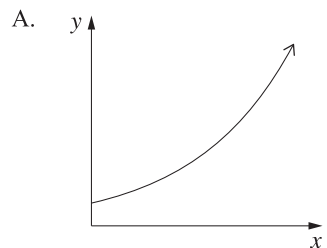
Find all possible values of m . (2 marks)

51. Functions, 2ADV F2 2013 HSC 15c

-  Part i: RAP Data (vs state mean) - Bottom 18%: School result (42%) was 7% below state average (49%)
-  Part ii: RAP Data (vs state mean) - Bottom 25%: School result (21%) was 4% below state average (25%)
-  Part ii: RAP Data (raw) - School result was 21%

i. Sketch the graph $y = |2x - 3|$. (1 mark)

ii. Using the graph from part (i), or otherwise, find all values of m for which the equation $|2x - 3| = mx + 1$ has exactly one solution. (2 marks)

Which graph could represent $y = 4^x$?**Worked Solutions**

1. Functions, 2ADV F2 2016 HSC 3 MC

$$y = 3 - (x-2)^2$$

Curve is concave down and passes through $(2, 3)$ $\Rightarrow B$

2. Functions, 2ADV F2 EQ-Bank 2 MC

Transforming $y = x^2$:Translate 1 unit left $\Rightarrow y = (x + 1)^2$ Dilate from x -axis by a factor of 2 $\Rightarrow y = 2(x + 1)^2$ Reflect in x -axis $\Rightarrow y = -2(x + 1)^2$ Translate 1 unit up $\Rightarrow y = 1 - 2(x + 1)^2$ $\therefore$ Transformations describe graph A. $\Rightarrow A$

3. Functions, 2ADV F2 2020 HSC 2 MC

Horizontal translation: 2 units to the right

$$x^3 \rightarrow (x-2)^3$$

Vertical translation: 5 units up

$$(x-2)^3 \rightarrow (x-2)^3 + 5$$

 $\Rightarrow B$

4. Functions, 2ADV F2 SM-Bank 9 MC

$$\begin{aligned} f(x) &= \frac{3x+2}{5-x} \\ &= \frac{-(15-3x)+17}{5-x} \\ &= -3 + \frac{17}{5-x} \end{aligned}$$

Vertical asymptote: $x = 5$ Horizontal asymptote: $y = -3$ $\Rightarrow D$

5. Functions, 2ADV’ F2 2019 HSC 4 MC

By elimination:

Graph is an even function

$\Rightarrow f(x) = f(-x)$

$\therefore$ Eliminate A and C

When $-1 < x < 1$, $y \leq 0$

$\therefore$ Eliminate D

$\Rightarrow B$

6. Functions, 2ADV F2 SM-Bank 5 MC

1st transformation:

$P(4, -3) \rightarrow (4, 1)$

2nd transformation:

$(4, 1) \rightarrow (-4, 1)$

$\Rightarrow A$

7. Functions, 2ADV F2 2014 HSC 2 MC

$y = (x - 1)^2 \rightarrow$ vertex $(1, 0)$

$\Rightarrow B$

8. Functions, 2ADV F2 2021 HSC 5 MC

By elimination:

When $x = 0$, $y = 10(0.8)^0 = 10$

$\rightarrow$ Eliminate B and D

As $x \rightarrow \infty$, $y \rightarrow 0$

$\rightarrow$ Eliminate C

$\Rightarrow A$

9. Functions, 2ADV F2 2025 HSC 6 MC

$y = -f(x) \Rightarrow$ Reflect $f(x)$ in the x -axis.

$y = -f(-x) \Rightarrow$ Reflect $-f(x)$ in the y -axis.

In this combination of translations, the order is not important.

$\Rightarrow C$

10. Functions, 2ADV F2 EQ-Bank 3 MC

Translate 3 units to the left:

$y = \frac{2}{x-2} \Rightarrow y' = \frac{2}{(x+3)-2} = \frac{2}{x+1}$

Dilate vertically by a factor of 2:

$y' = \frac{2}{x+1} \Rightarrow \frac{y''}{2} = \frac{2}{x+1}$

$\Rightarrow D$

11. Functions, 2ADV F2 SM-Bank 6 MC

Let $y = \sqrt{2x-5}$

1st transformation:

$y = -\sqrt{2x-5}$

2nd transformation:

Dilate from y -axis by a factor of $\frac{1}{2}$

$\Rightarrow$ Swap $x \rightarrow 2x$

$y = -\sqrt{2(2x)-5}$

$= -\sqrt{4x-5}$

$\therefore f(x) = -\sqrt{4x-5}$

$\Rightarrow D$

COMMENT: Using “swap” terminology for dilations from the y -axis is simpler and more intelligible for students in our view.

12. Functions, 2ADV F2 SM-Bank 4 MC

Let $y = 3x^{\frac{5}{2}}$

Reflect in the x -axis:

$y = -3x^{\frac{5}{2}}$

Translate 3 units to the right:

$y = -3(x-3)^{\frac{5}{2}}$

Translate 4 units down:

$y = -3(x-3)^{\frac{5}{2}}-4$

$\Rightarrow B$

13. Functions, 2ADV F2 SM-Bank 10 MC

$$\begin{aligned} y &= -\left(\frac{3-x}{2-x}\right) \\ &= -\left(\frac{2-x}{2-x} + \frac{1}{2-x}\right) \\ &= -1 - \frac{1}{2-x} \end{aligned}$$

∴ Asymptotes: $x = 2, y = -1$
⇒ C

14. Functions, 2ADV’ F2 2015 HSC 5 MC

$$y = \frac{3x}{(x+1)(x+2)}$$

Asymptotes at $x = -1$ and $x = -2$

As $x \rightarrow \infty, y \rightarrow 0^+$

As $x \rightarrow -\infty, y \rightarrow 0^-$

∴ Horizontal asymptote at $y = 0$
⇒ A

15. Functions, 2ADV’ F2 2017 HSC 5 MC

$$\begin{aligned} y &= \frac{2x^2}{(1-x^2)} \\ &= -\frac{(2-2x^2-2)}{(1-x^2)} \\ &= -\frac{2(1-x^2)}{(1-x^2)} - \frac{2}{(1-x^2)} \\ &= -2 - \frac{2}{(1-x^2)} \end{aligned}$$

As $x \rightarrow \infty, y \rightarrow -2$
∴ Horizontal asymptote at $y = -2$
⇒ D

16. Functions, 2ADV F2 2024 HSC 4 MC

$$y = (x-3)^2 - 2$$

Reflect in the y -axis ($f(-x)$):

$$y = (-x-3)^2 - 2$$

Reflect in the x -axis ($-f(-x)$):

$$\begin{aligned} y &= -[(-x-3)^2 - 2] \\ &= -(x+3)^2 + 2 \end{aligned}$$

⇒ C

17. Functions, 2ADV F2 2024 HSC 7 MC

At $x = 0$:

$$f(2x-1) = f(-1) \Rightarrow \text{Eliminate A and B.}$$

Consider the transformations of $f(x) \rightarrow f(2x-1)$

→ Shift $f(x)$ 1 unit to the right.

→ Dilate $f(x-1)$ by a factor of $\frac{1}{2}$ from the y -axis.

⇒ C

18. Functions, 2ADV F2 2024 MET2 12 MC

By elimination:

Graph has been dilated by a factor of $\frac{1}{2}$ from y -axis.

Eliminate C and D.

Graph is then translated $\frac{1}{2}$ unit to the left.

Consider the turning point $(2, 1)$ after translation:

$$(2, 1) \rightarrow \left(2 \times \frac{1}{2} - \frac{1}{2}, 1\right) = \left(\frac{1}{2}, 1\right)$$

∴ Option A is the only possible solution.

⇒ A

19. Functions, 2ADV F2 2022 SPEC2 3 MC

$$y = \frac{x^2 + 2x + c}{x^2 - 4} \Rightarrow y = \frac{x^2 + 2x + c}{(x - 2)(x + 2)}$$

Vertical asymptotes: $x = 2$, $x = -2$

→ However, if $c = 0$, only 1 vertical asymptote at $x = 2$.

Horizontal asymptote: $y = 1$

⇒ *D*

Mean mark 38%.
41% of students chose
A and did not consider
when $c = 0$

20. Functions, 2ADV F2 SM-Bank 7 MC

$$g(x) = \frac{1}{2}f(x-1), \quad A(3, 2)$$

Dilate by a factor of $\frac{1}{2}$ from x -axis:

$$A(3, 2) \rightarrow A'(3, 1)$$

Translate 1 unit to right:

$$A'(3, 1) \rightarrow P(4, 1)$$

⇒ *C*

21. Functions, 2ADV F2 SM-Bank 8 MC

$$\text{Let } f(x) = \sqrt{8x^3 + 1}$$

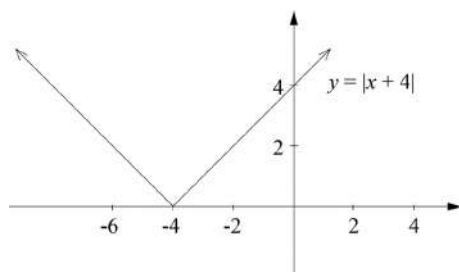
$$\begin{aligned} f\left(\frac{1}{2}x\right) &= \sqrt{8\left(\frac{1}{2}x\right)^3 + 1} \\ &= \sqrt{x^3 + 1} \end{aligned}$$

∴ Transformation correct when x is swapped for $\frac{x}{2}$

i.e. graph is dilated by factor of 2 from y -axis

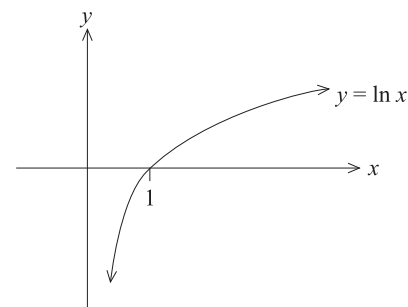
⇒ *A*

22. Functions, 2ADV F2 2006 HSC 1c



23. Functions, 2ADV F2 SM-Bank 1

i.

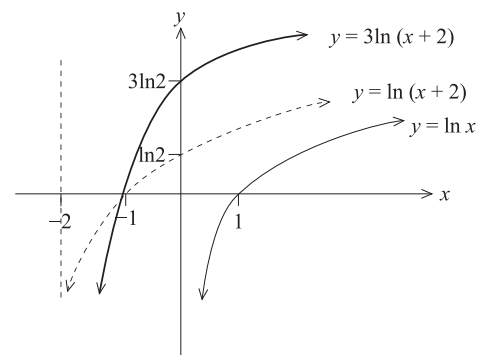


ii. Transforming $y = \ln x \Rightarrow y = \ln(x + 2)$

$y = \ln x \Rightarrow$ shift 2 units to left.

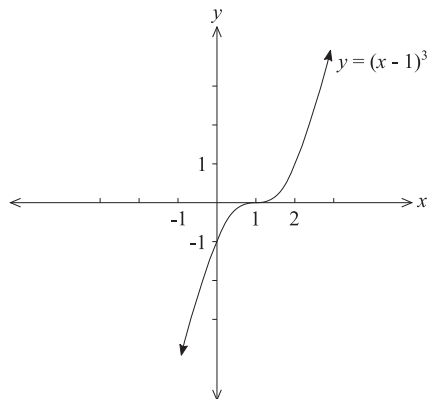
Transforming $y = \ln(x + 2)$ to $y = 3\ln(x + 2)$

⇒ increase each y value by a factor of 3

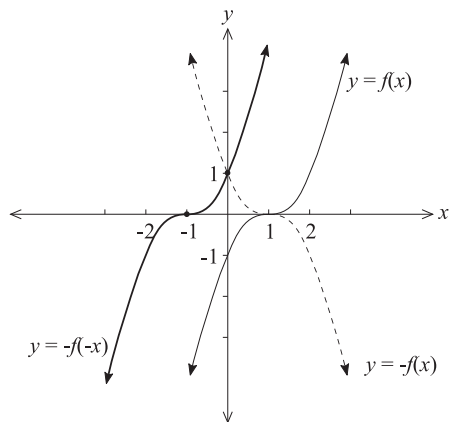


24. Functions, 2ADV F2 SM-Bank 35

- i. $y = (x-1)^3 \Rightarrow y = x^3$ shifted 1 unit to the right.



- ii. $y = -f(x) \Rightarrow$ reflect $y = (x-1)^3$ in x -axis.
 $y = -f(-x) \Rightarrow$ reflect $y = -f(x)$ in y -axis.



25. Functions, 2ADV F2 2023 HSC 19

- a. $g(x)$ cuts x -axis at 1 and -3 .

$$g(x)_{\max} = g(-1) = 4$$

Find intersection of graphs:

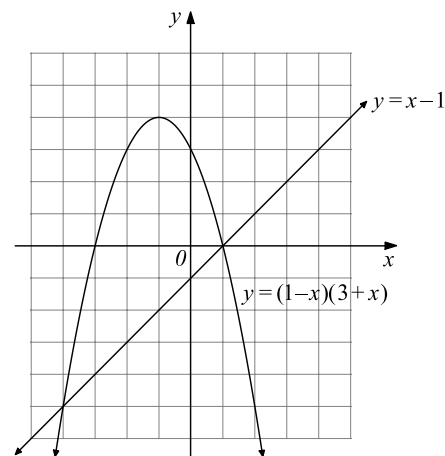
$$(1-x)(3+x) = x-1$$

$$3+x-3x-x^2x-1$$

$$x^2+3x-4=0$$

$$(x+4)(x-1)=0$$

Intersections at: $(1, 0), (-4, -5)$



- b. From the graph:

$$x-1 < (1-x)(3+x) \text{ when } -4 < x < 1$$

Test $x = 0$:

$$0-1 < (1-0)(3+0) \Rightarrow -1 < 3 \text{ (correct)}$$

26. Functions, 2ADV F2 EQ-Bank 16

i. Transformation 1:

Translate $y = x^4$ 2 units to the left.

$$y = x^4 \Rightarrow y = (x + 2)^4$$

Transformation 2:

Dilate $y = (x + 2)^4$ by a factor of $\frac{1}{3}$ from the x -axis

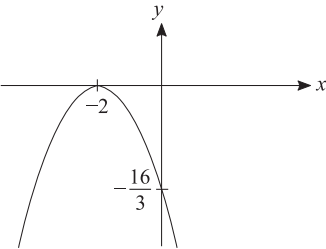
$$y = (x + 2)^4 \Rightarrow y = \frac{(x + 2)^4}{3}$$

Transformation 3:

Reflect $y = \frac{(x + 2)^4}{3}$ in the x -axis.

$$y = \frac{(x + 2)^4}{3} \Rightarrow y = -\frac{(x + 2)^4}{3}$$

ii.



27. Functions, 2ADV F2 EQ-Bank 9

i. $4x-1 \neq 0$

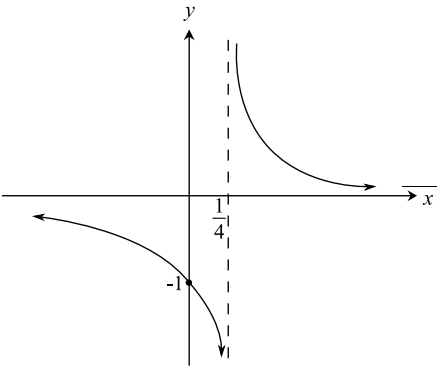
$$x \neq \frac{1}{4}$$

$$\therefore \text{Domain: } \left\{ \text{all real } x, x \neq \frac{1}{4} \right\}$$

ii. When $x = 0, y = -1$

As $x \rightarrow \infty, y \rightarrow 0^+$

As $x \rightarrow -\infty, y \rightarrow 0^-$



28. Functions, 2ADV' F2 2012 HSC 13b

$$\begin{aligned} \text{i. } y &= \frac{2x^2}{x^2 + 9} \\ &= \frac{2}{1 + \frac{9}{x^2}} \end{aligned}$$

As $x \rightarrow \infty$, $y \rightarrow 2$

As $x \rightarrow -\infty$, $y \rightarrow 2$

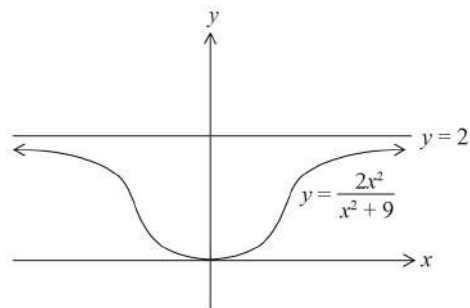
$\therefore$ Horizontal asymptote at $y = 2$

ii. At $x = 0$, $y = 0$

$$f(x) = \frac{2x^2}{x^2 + 9} \geq 0 \text{ for all } x$$

$$f(-x) = \frac{2(-x)^2}{(-x)^2 + 9} = \frac{2x^2}{x^2 + 9} = f(x)$$

Since $f(x) = f(-x) \Rightarrow$ EVEN function



29. Functions, 2ADV F2 2021 HSC 19

Asymptotes: $x = -4$

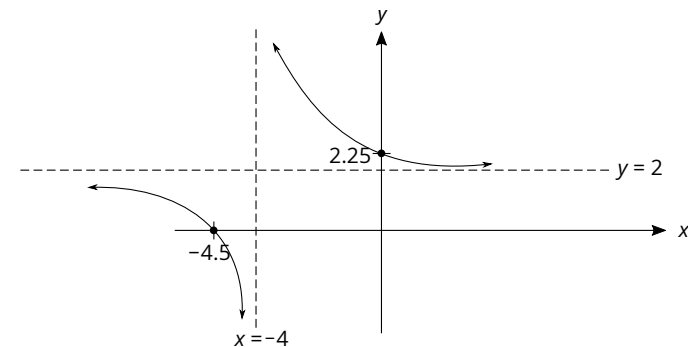
As $x \rightarrow \infty$, $y \rightarrow 2$

y -intercept occurs when $x = 0$:

$$y = 2.25$$

x -intercept occurs when $y = 0$:

$$2 + \frac{1}{x + 4} = 0 \Rightarrow x = -4.5$$



30. Functions, 2ADV F2 2024 MET1 3*

Find asymptotes:

$(x + 3) \neq 0 \Rightarrow$ Asymptote at $x = -3$

As $x \rightarrow \infty$, $\frac{1}{(x + 3)^2} \rightarrow 0 \Rightarrow$ Asymptote at $y = -2$

y -intercept: $x = 0$

$y = \frac{1}{(0 + 3)^2} - 2 = -\frac{17}{9}$

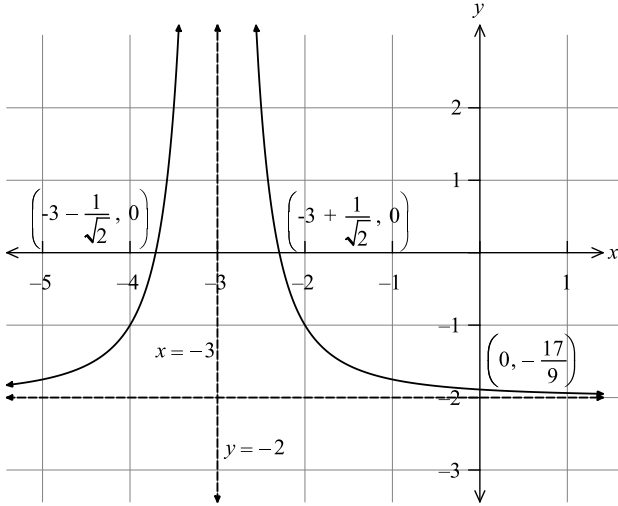
x -intercepts: $y = 0$

$\frac{1}{(x + 3)^2} - 2 = 0$

$(x + 3)^2 = \frac{1}{2}$

$x + 3 = \pm \frac{1}{\sqrt{2}}$

$x = -3 \pm \frac{1}{\sqrt{2}}$



31. Functions, 2ADV F2 2025 HSC 30

$y = (x - 1)(x - 5)$

Translate k units to the right:

$y \rightarrow y' = (x - k - 1)(x - k - 5)$

Translate k units vertically up:

$y' \rightarrow y'' = (x - k - 1)(x - k - 5) + k$

y'' passes through $(6, 11)$:

$11 = (6 - k - 1)(6 - k - 5) + k$

$11 = (5 - k)(1 - k) + k$

$11 = 5 - 6k + k^2 + k$

$0 = k^2 - 5k - 6$

$0 = (k - 6)(k + 1)$

$\therefore k = 6 \quad (k > 0)$

32. Functions, 2ADV F2 EQ-Bank 1

$y = |x|$

Translate 3 units left $\Rightarrow y = |x + 3|$

Reflect in the x -axis $\Rightarrow y = -|x + 3|$

Dilate by $\frac{1}{3}$ from the x -axis

$\Rightarrow$ Multiply by $\frac{1}{3} \Rightarrow y = -\frac{1}{3}|x + 3|$

Translate 2 units up $\Rightarrow y = -\frac{1}{3}|x + 3| + 2$

$\therefore k = -\frac{1}{3}, b = 3, c = 2$

33. Functions, 2ADV F2 EQ-Bank 14

$$y = x^2$$

Transformation 1:

Translate 3 units in negative x -direction

$$y = (x + 3)^2$$

$$y = x^2 + 6x + 9$$

Transformation 2:

Translate 10 units in negative y -direction

$$y = x^2 + 6x - 1$$

Transformation 3:

Reflect in the x -axis

$$y = - (x^2 + 6x - 1)$$

$$= 1 - 6x - x^2$$

34. Functions, 2ADV F2 EQ-Bank 2

Vertical asymptote at $x = -1 \Rightarrow p = 1$

Horizontal asymptote at $y = \frac{3}{2} \Rightarrow q = \frac{3}{2}$

$$y = \frac{1}{x + 1} + \frac{3}{2}$$

x -intercept when $y = 0$:

$$\frac{1}{x + 1} + \frac{3}{2} = 0$$

$$\frac{1}{x + 1} = -\frac{3}{2}$$

$$3x + 3 = -2$$

$$3x = -5$$

$$x = -\frac{5}{3}$$

35. Functions, 2ADV F2 SM-Bank 16

1st transformation

Dilation by a factor of $\frac{1}{2}$ from the y -axis:

$$x^2 - 4 \Rightarrow \left(\frac{x}{\frac{1}{2}}\right)^2 - 4 = 4x^2 - 4$$

2nd transformation

Translation by 2 units to the right:

$$4x^2 - 4 \Rightarrow g(x) = 4(x - 2)^2 - 4$$

x -axis intercept of $g(x)$:

$$4(x - 2)^2 - 4 = 0$$

$$(x - 2)^2 = 1$$

$$x - 2 = \pm 1$$

$$x - 2 = 1 \Rightarrow x = 3$$

$$x - 2 = -1 \Rightarrow x = 1$$

$\therefore$ Horizontal axis intercepts occur at $(1, 0)$ and $(3, 0)$.

36. Functions, 2ADV F2 SM-Bank 13

i. $f(x) = \frac{1-e^x}{1+e^x}$

$$\begin{aligned} f(-x) &= \frac{1-e^{-x}}{1+e^{-x}} \times \frac{e^x}{e^x} \\ &= \frac{e^x-1}{e^x+1} \\ &= -\frac{1-e^x}{1+e^x} \\ &= -f(x) \end{aligned}$$

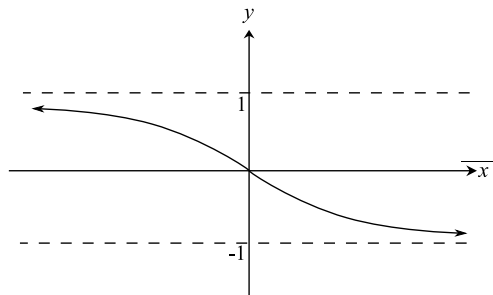
$\therefore f(x)$ is ODD.

ii. $y = \frac{1-e^x}{1+e^x} \times \frac{e^{-x}}{e^{-x}} = \frac{e^{-x}-1}{e^{-x}+1} = 1 - \frac{2}{e^{-x}+1}$

As $x \rightarrow \infty$, $\frac{2}{e^{-x}+1} \rightarrow 2$, $y \rightarrow -1$

As $x \rightarrow -\infty$, $\frac{2}{e^{-x}+1} \rightarrow 0$, $y \rightarrow 1$

When $x = 0$, $y = 0$

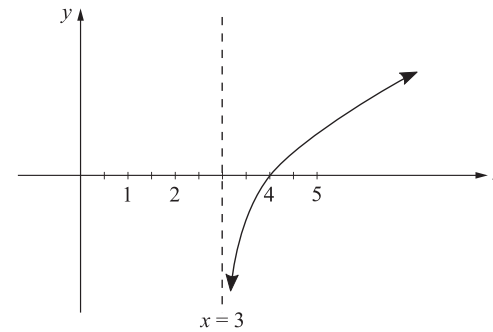


37. Functions, 2ADV F2 SM-Bank 2

Asymptote when $x = 3$

Domain $\{x: x > 3\}$

Range: all y



38. Functions, 2ADV F2 SM-Bank 3

i. Vertical Asymptote:

$$x = -1$$

$$\therefore a = 1$$

ii. Solve $f(0) = 1$ for c ,

$$c = 1$$

iii. $f(x) = k \log_e(x+1) + 1$

Since $f(p) = 10$,

$$k \log_e(p+1) + 1 = 10$$

$$k \log_e(p+1) = 9$$

$$\therefore k = \frac{9}{\log_e(p+1)} \dots \text{as required}$$

39. Functions, 2ADV F2 SM-Bank 10

i. $4 - x^2 \neq 0$
 $\Rightarrow x \neq 2$ or -2
 $\therefore$ Domain: all x , $x \neq \pm 2$

ii. Asymptotes at $x = \pm 2$

Passes through $(0, 0)$

As $x \rightarrow 2^-$, $y \rightarrow \infty$

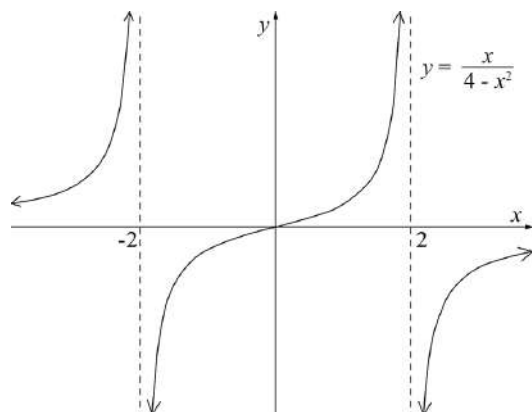
$x \rightarrow 2^+$, $y \rightarrow -\infty$

As $x \rightarrow -2^-$, $y \rightarrow \infty$

$x \rightarrow -2^+$, $y \rightarrow -\infty$

As $x \rightarrow \infty$, $y \rightarrow 0$

$x \rightarrow -\infty$, $y \rightarrow 0$



COMMENT: Note that $x \rightarrow 2^-$ is a short way of writing as x approaches 2 from the negative (or left-hand) side. This notation can save time when required.

40. Functions, 2ADV F2 EQ-Bank 11

$$\frac{2x - 2}{x + 1} = \frac{2(x + 1) - 4}{x + 1}$$

$$= 2 - \frac{4}{(x + 1)}$$

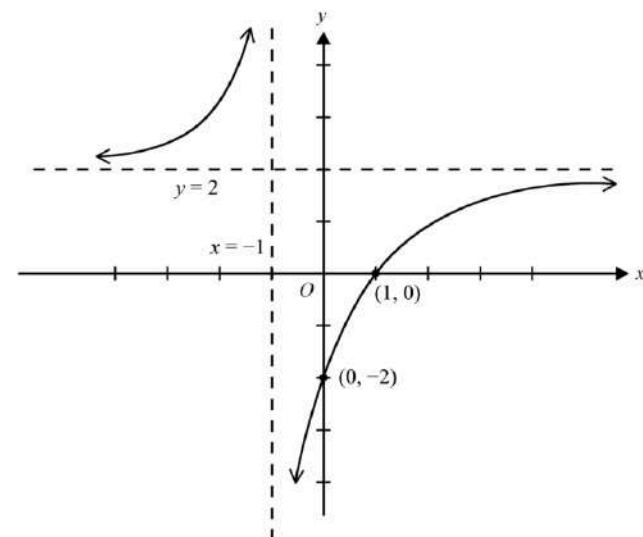
Asymptotes: $x = -1$, $y = 2$

As $x \rightarrow \infty$, $y \rightarrow 2(-)$

As $x \rightarrow -\infty$, $y \rightarrow 2(+)$

As $x \rightarrow -1(-)$, $y \rightarrow \infty$

As $x \rightarrow -1(+)$, $y \rightarrow -\infty$



41. Functions, 2ADV F2 SM-Bank 12

$$\begin{aligned}\frac{2x+1}{x-1} &= \frac{2x-2+3}{x-1} \\ &= \frac{2(x-1)+3}{x-1} \\ &= 2 + \frac{3}{x-1}\end{aligned}$$

COMMENT:

Manipulation of the equation makes graphing much easier.

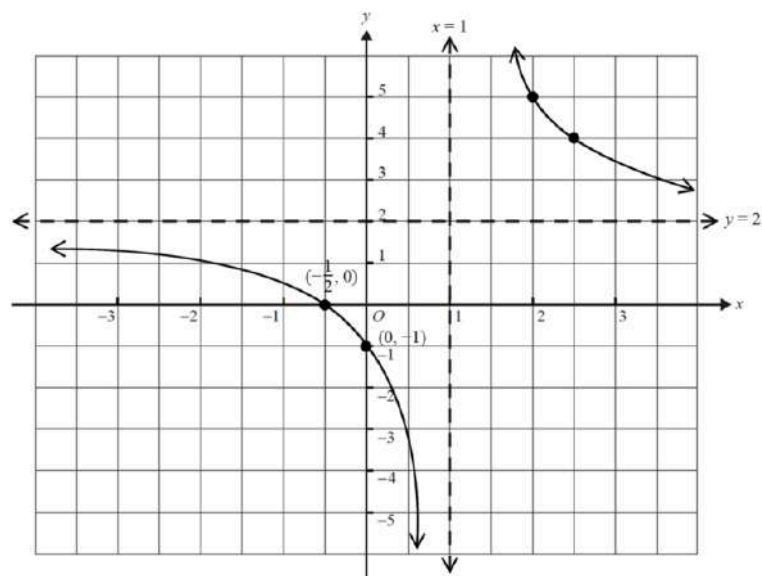
Asymptotes: $x = 1$, $y = 2$

As $x \rightarrow \infty$, $y \rightarrow 2(+)$

As $x \rightarrow -\infty$, $y \rightarrow 2(-)$

As $x \rightarrow -1(-)$, $y \rightarrow -\infty$

As $x \rightarrow -1(+)$, $y \rightarrow \infty$



42. Functions, 2ADV' F2 2007 HSC 3b

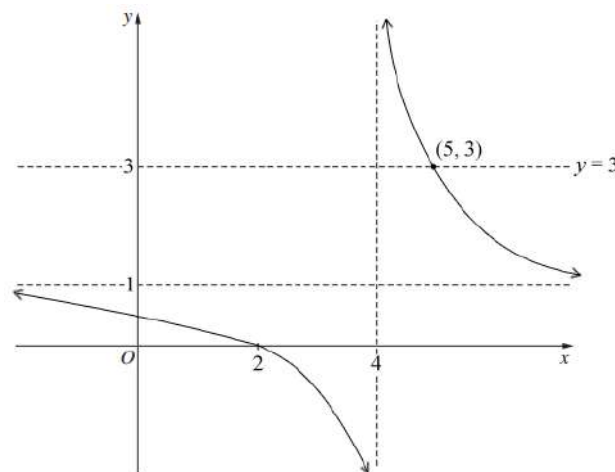
i. $y = \frac{x-2}{x-4}$

Vertical asymptote at $x = 4$

$$\lim_{x \rightarrow \infty} \frac{x-2}{x-4} = \lim_{x \rightarrow \infty} \frac{1 - \frac{2}{x}}{1 - \frac{4}{x}} = 1$$

$$y\text{-intercept} = \frac{1}{2}$$

$$x\text{-intercept} = 2$$



ii. Find x so that $\frac{x-2}{x-4} \leq 3$

$$\frac{x-2}{x-4} = 3$$

$$x-2 = 3x-12$$

$$2x = 10$$

$$x = 5$$

$\Rightarrow (5, 3)$ is the intersection of $y = 3$ and $y = \frac{x-2}{x-4}$

$\therefore \frac{x-2}{x-4} \leq 3$ when $x < 4$ and $x \geq 5$.

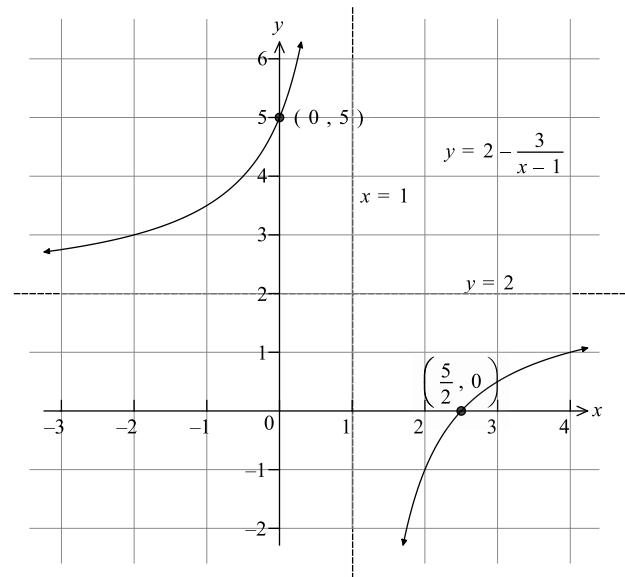
43. Functions, 2ADV F2 2023 MET1 3

- a. Vertical asymptote when $x = 1$

$$y\text{-int: } y = 2 - (-3) \Rightarrow y = 5$$

$$x\text{-int: } 2 - \frac{3}{x-1} = 0 \Rightarrow x = \frac{5}{2}$$

$$\text{As } x \rightarrow \infty, y \rightarrow 2^-; x \rightarrow -\infty, y \rightarrow 2^+$$



- b. From the graph:

$$f(x) = 1 \text{ when } x = 4$$

$x > 1$ to the right of the vertical asymptote

$$\therefore f(x) \leq 1 \text{ when } 1 < x \leq 4 \text{ or } (1, 4]$$

Mean mark (b) 38%.
MARKER'S COMMENT:
Many students did not use their graph from part (a).

44. Functions, 2ADV F2 EQ-Bank 13

i. $y = kx^2 + c$

Translate 2 units in positive x -direction.

$$y = kx^2 + c \Rightarrow y = k(x-2)^2 + c$$

Dilate in the positive y -direction by a factor of 4.

$$y = k(x-2)^2 + c \Rightarrow y = 4k(x-2)^2 + 4c$$

ii. $y = 4k(x^2 - 4x + 4) + 4c$

$$= 4kx^2 - 16kx + 16k + 4c$$

Reflect in the y -axis.

$$\Rightarrow \text{Swap: } x \rightarrow -x$$

$$y = 4k(-x)^2 - 16k(-x) + 16k + 4c$$

$$= 4kx^2 + 16kx + 16k + 4c$$

Equating co-efficients:

$$4k = 8$$

$$\therefore k = 2$$

$$16k + 4c = -8$$

$$4c = -40$$

$$\therefore c = -10$$

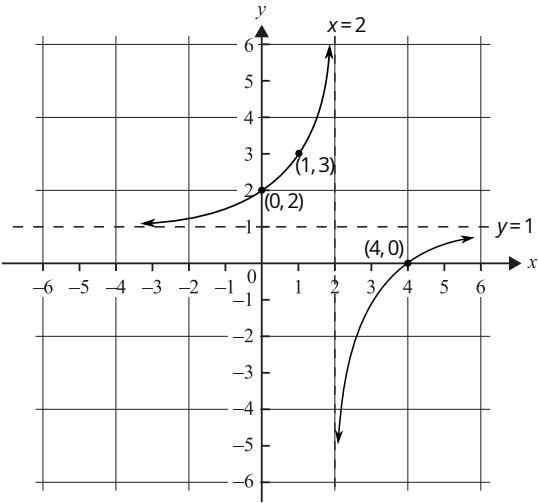
COMMENT: Using "swap" terminology for reflections in the y -axis is simpler and more intelligible for students in our view.

45. Functions, 2ADV F2 EQ-Bank 2

- a. $g(x) = 3[2(x + 2)]^2$
- $$= 3 \times 4(x + 2)^2$$
- $$= 12(x + 2)^2$$
- b. $P(-3, 9)$ lies on $f(x) = x^2$
- Find corresponding point on $f(x)$
- Mapping x_f to x_g :
- $$2(x_g + 2) = x_f \Rightarrow x_g = \frac{1}{2}x_f - 2$$
- $$x_g = \frac{1}{2} \times -3 - 2 = -\frac{7}{2}$$
- Mapping y_f to y_g :
- $$y_g = 3 \times y_f = 3 \times 9 = 27$$
- $\therefore$ Corresponding point $= \left(-\frac{7}{2}, 27\right)$
-

46. Functions, 2ADV F2 SM-Bank 20

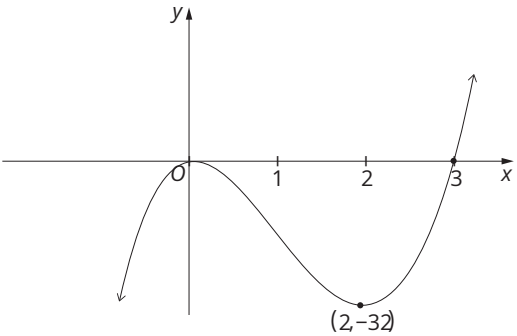
- a. Asymptotes:
- $x = 2$
- As $x \rightarrow \pm \infty, y \rightarrow 1 \Rightarrow$ Asymptote at $y = 1$
- y -intercept at $(0, 2)$
- x -intercept at $(4, 0)$



- b. By inspection of the graph:

$$1 - \frac{2}{x - 2} \geq 3 \text{ for } x \in [1, 2)$$

47. Functions, 2ADV F2 2021 HSC 21



Mean mark 48%.

48. Functions, 2ADV F2 SM-Bank 2

$f(x) = (x + 2)^2 \Rightarrow f(x + a) = (x + a + 2)^2$

Horizontal translation: 3 units to right

$y = kf(x - 3) + c$

Vertical translation: 2 units up

$y = kf(x - 3) + 2$

Since (0, 3) lies on the transformed function:

$3 = kf(-3) + 2$

$3 = k + 2$

$k = 1$

$\therefore a = -3, c = 2, k = 1$

49. Functions, 2ADV F2 2022 HSC 19

Horizontal translation m units to the right:

$x^2 \rightarrow (x - m)^2$

Dilated vertically by scale factor k :

$(x - m)^2 \rightarrow k(x - m)^2$

Vertical translation 5 units down:

$k(x - m)^2 \rightarrow k(x - m)^2 - 5$

$y = k(x - m)^2 - 5$

$= k(x^2 - 2mx + m^2) - 5$

$= kx^2 - 2kmx + (km^2 - 5)$

$\therefore k = 3$

$-2km = -12$

$\therefore m = 2$

Mean mark 51%.

50. Functions, 2ADV F2 2023 HSC 27

- a. Consider the transformation of $y = -|x|$

Translate 6 units to the right

$y = -|x| \rightarrow y = -|x - 6|$

$\therefore b = 6$

Translate 7 units vertically up

$y = -|x - 6| \rightarrow y = -|x - 6| + 7$

$\therefore c = 7$

$f(x) = a|x - 6| + 7$ passes through (3, -5):

$-5 = a|3 - 6| + 7$

$-5 = 3a + 7$

$3a = -12$

$\therefore a = -4$

Mean mark (b) 23%.

- b. $y = mx$ passes through (0, 0)

One solution when $y = mx$ passes through (0, 0) and (6, 7)

$m = \frac{7 - 0}{6 - 0} = \frac{7}{6}$

As graph gets flatter and turns negative $\Rightarrow$ 2 solutions

2 solutions continue until $y = mx$ is parallel to

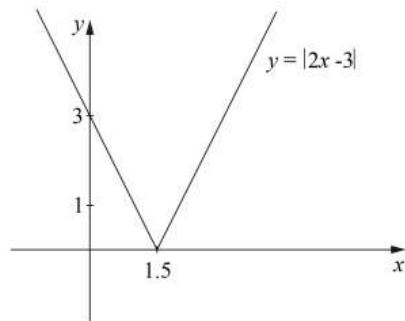
the line joining (6, 7) to (9, -5), where:

$m = \frac{7 - (-5)}{6 - 9} = -\frac{12}{3} = -4$

$\therefore$ 2 solutions when $-4 < m < \frac{7}{6}$

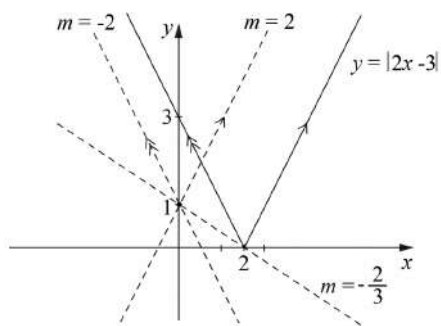
51. Functions, 2ADV F2 2013 HSC 15c

i.



Mean mark 49%
MARKER'S COMMENT: Many students drew diagrams that were "too small", didn't use rulers or didn't use a consistent scale on the axes!

ii.



Line of intersection $y = mx + 1$ passes through $(0, 1)$

If it also passes through $(1.5, 0) \Rightarrow 1$ solution

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{1 - 0}{0 - \frac{3}{2}} \\ &= -\frac{2}{3} \end{aligned}$$

Mean mark 25%.
COMMENT: Students need a clear graphical understanding of what they are finding to solve this very challenging, Band 6 question.

Gradients of $y = |2x - 3|$ are 2 or -2

Considering a line through $(0, 1)$:

If $m \geq 2$, only intersects once.

Similarly,

If $m < -2$, only intersects once.

$\therefore$ Only one solution when $m = -\frac{2}{3}$, $m \geq 2$ or $m < -2$

52. Functions, 2ADV F2 2025 HSC 2 MC

$\Rightarrow A$

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